

EUROTCL 2026

undroidwish for Apple Silicon Macs

Now with bluetooth and borg compatibility

github.com/johnbuckman/undroidwish-arm64-batteries-included · under the Tcl/Tk license

John Buckman EuroTcl 2026 · Vienna

WHY NOW

Rosetta 2 is being retired

● Jun 2020	WWDC: the Apple Silicon switch begins. Rosetta 2 is billed as transitional.
● Nov 2020	First M1 Macs. Intel apps run only via Rosetta 2 translation.
● Jun 2025	WWDC: Rosetta 2 promised for only two more releases — through macOS 27.
● Feb 2026	macOS 26.4 starts warning users about Rosetta-dependent apps.
● Fall 2026	macOS 27 “Golden Gate” — last release with full Rosetta 2.
● Fall 2027	macOS 28 removes Rosetta 2 — x86 apps stop launching.



Living on borrowed time

An x86 undroidwish only runs through a translation layer Apple is deleting. A native arm64 build is the only way de1app keeps running on macOS 27 and beyond.

BENCHMARKS

Native arm64 vs x86 under Rosetta 2

1.46×

faster overall

geometric mean of all micro-benchmarks
median of 5 runs · M4 MacBook Air

Startup

1.25× 20%



Math

1.47× 47%



General Tcl

1.53× 53%



Tk

1.31× 31%



Tk gains are smallest because much of that work is SDL surface blitting on the GPU — CPU-heavy math and list work shows the biggest native win.

BENCHMARKS

Every metric — x86 vs arm64 (ms)

metric	x86	arm64	faster	metric	x86	arm64	faster
startup	531.7	425.0	1.25×	list.build_sort	137.7	83.1	1.66×
math.int	954.1	583.4	1.64×	list.foreach_search	125.3	59.2	2.12×
math.float	894.7	623.0	1.44×	dict.set_get	396.5	265.7	1.49×
math.trig	519.7	336.4	1.54×	regexp.match	1258.9	982.5	1.28×
math.fib_recursion	422.6	281.4	1.50×	array.set_get	157.7	110.0	1.43×
math.bignum	44.6	34.9	1.28×	binary.format_scan	167.7	109.1	1.54×
str.append	47.4	28.9	1.64×	tk.widget_create	20.6	15.5	1.33×
str.map_regsub	320.2	260.0	1.23×	tk.canvas_items	12.0	9.5	1.26×
str.format	284.0	184.1	1.54×	tk.canvas_redraw	124.3	95.4	1.30×
				tk.text_insert	4.3	3.2	1.34×

01

The borg command

AndroWish's platform bridge — reintroduced on the Mac desktop



BACKGROUND

What borg does

- On AndroWish, borg exposes the Android platform to Tcl
- Notifications, speech, USB, preferences, device info
- One command, dozens of subcommands — a native API in Tcl
- de1app is written against borg, assuming Android underneath
- But undroidwish ships “sans the borg” — it's missing



borg

The platform bridge de1app expects — so we bring it back on the Mac.

IMPLEMENTATION

tkBorgOSX.c — a clean-room port

- Not an `#ifdef`-shred of the 6,900-line Android JNI file
- A self-contained reimplementation guarded by `__APPLE__`
- Exposes the same subcommand surface — code runs unmodified
- Never errors on a well-formed call; absent features no-op
- Built only on libc, Tcl, CoreFoundation, IOKit, CoreGraphics



`package require Borg`

Ships as a loadable Tcl stubs package, bundled into `assets.zip` — the same way undroidwish carries all its batteries.

The same calls, native backends

```
borg.tcl

# the same delapp code, unmodified
borg speak "Shot ready"
borg toast "Saved"
borg beep

set bi [borg osbuildinfo]
# manufacturer -> Apple
# model        -> Mac16,12

borg usbdevices
```

- Native backends, Android-shaped output
- say for speech, afplay for beep
- An always-foreground SDL panel for toast
- IOKit enumerates USB; DisplayServices sets brightness

What works natively on the Mac



Speech & sound

speak · stopspeak · beep · toast — via say, afplay and an always-foreground SDL overlay.



Device & display

osbuildinfo · displaymetrics · brightness · networkinfo — real values from sysctl & CoreGraphics.



Storage & prefs

sharedpreferences · osenvironment — a typed KV store under ~/Library/Application Support.



USB & app control

usbdevices · withdraw · spinner · log · locale — IOKit enumeration and native window control.

Graceful where macOS differs



Partial — common case covered

- activity → open a URL or file
- notification → osascript banners
- bluetooth state via system_profiler
- permissions report granted



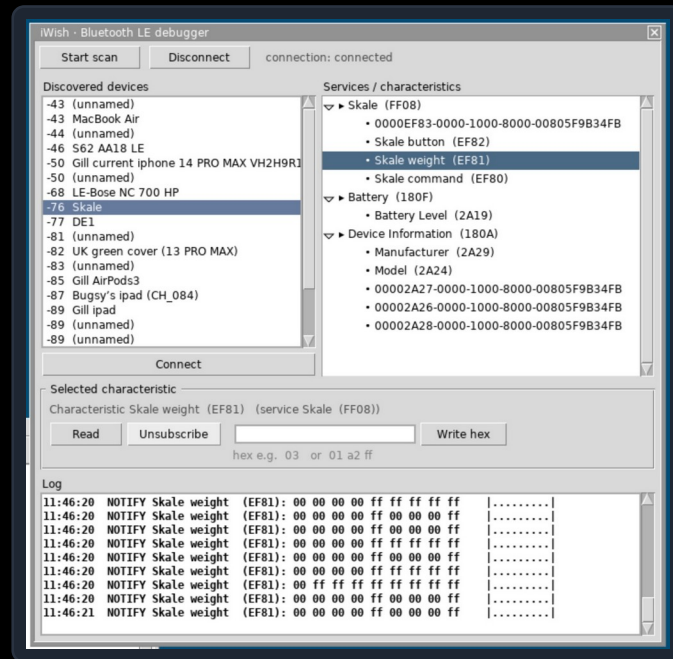
Stub — accepted, never errors

- vibrate · sms · NFC — no hardware
- camera · location · sensors — future work
- Return Android-shaped empty values
- So de1app runs without a single crash

02

Bluetooth LE on the Mac

The Android ble command set, now driven by CoreBluetooth



One ble API, CoreBluetooth underneath

- **undroidwish now bundles a native BLE driver for macOS**
- It exposes the exact same ble command set as AndroWish
- scan · connect · discover · read · write · subscribe
- Built on CoreBluetooth, marshalled onto the Tcl event loop
- de1app's Android BLE code runs unchanged on the Mac



```
package require ble
```

Completely compatible with the Android ble command set — write the BLE code once, run it on both.

One Tcl/Tk code base running on Mac, iPad and Android

- Native arm64 on Apple Silicon — signed and notarized
- borg and ble reintroduced, with native Apple backends
- Mac Desktop, iOS and Android from a single Tcl script